orthrus Credential Broker — Linux Kerberos CLIENT ↔ Active Directory KDC

The broker is an MIT krb5 CLIENT of AD (no KDC of its own); it mints per-user service tickets by constrained delegation (S4U) across the realm edge

(the broker runs in S4U client mode against AD — not Sybil's native KDC-adjacent design; AD is the sole KDC)

WINDOWS BOUNDARY • Active Directory domain

CAC / PIV smartcard

User Windows workstation (AD-joined)

- CAC logon → PKINIT → user TGT (held locally)
- the user TGT stays on Windows

✗ **SSH GSSAPI credential forwarding is BLOCKED**

PKINIT (CAC)
→ user TGT

Active Directory Domain Controller — KDC

- AS / TGS (Kerberos, TCP 88)
- kadmin • LDAP • DNS
- PAC signing & validation (KB5008380)

Delegation policy (authorizes the broker's S4U)

- msDS-AllowedToDelegateTo = { hive/..., hdfs/... }
  ← the S4U allow-list (the boundary of trust)
- broker: TrustedToAuthForDelegation (protocol transition)
- users: delegation-eligible (not Protected Users)

user TGT not forwarded
(SSH GSSAPI blocked)
— why the broker exists

Kerberos • TCP 88

① **kinit -k → broker TGT**
(broker keytab)

② **S4U2Self → evidence ticket**

③ **S4U2Proxy → Hive/HDFS tkts**
(for <user>)

service ticket /
PAC validation

TRUST BOUNDARY

LINUX BOUNDARY • HPC cluster (UID / GID)

Login node — sshd / PAM

- CAC-gated OS login (UID / GID only)
- the node itself knows no Kerberos

krb-get (UNIX socket, GET)

orthrus — Kerberos client broker (root)

- MIT krb5 CLIENT of AD — NO KDC / realm of its own
  (libkrb5 / GSSAPI • broker keytab • krb5.conf • ccache)
- holds ONE broker keytab (gMSA / HSM-backable)
- broker TGT: kinit -k (renewable, unattended)
- mints per-user SERVICE tickets (S4U2Self + S4U2Proxy)
- installs as the user (setpriv → $HOME, 0600)
- front door: UNIX socket, SO_PEERCRED ◀ krb-get

write as user (setpriv)

Shared filesystem home
$HOME/.krb5/krb5cc_hpc
(0700 dir / 0600 file)
service tickets — no TGT

shared mount

Compute nodes cn001..cnNNN
Slurm TaskProlog →
KRB5CCNAME points here
(every node, same cache)

use service ticket (GSSAPI)

Kerberized backends — Hive / HDFS (AD realm)

- accept the user's hive/_HOST service ticket (GSSAPI)
- verify with own service key; check PAC for authz (Ranger)

Kerberos across boundary (TCP 88) · local / shared-filesystem path · blocked path (what the design works around) · validation · **Focal node: orthrus (amber)**